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INDR 252 CLASS EXERCISE, February 21, 2024

Consider the random variables X and Y with the following joint probability mass function:

$$P_{X,Y}(1,1) = P_{X,Y}(0,0) = \frac{3}{8}$$

$$P_{X,Y}(0,1) = P_{X,Y}(1,0) = \frac{1}{8}$$

$$P_{X,Y}(x,y) = 0 \text{ for all other } (x,y)$$

Compute the following quantities

- a) $E[X], E[Y]$
- b) $\text{Var}(X), \text{Var}(Y)$
- c) $\text{Cov}(X,Y)$
- d) $\text{Var}(2X+3Y)$

Note that:

$$\text{Var}(X) = E[(X - \mu_X)^2] = E[X^2] - (\mu_X)^2$$

$$\text{Cov}(X,Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X \mu_Y$$

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X,Y)$$

$$P_X(0) = P_{X,Y}(0,0) + P_{X,Y}(0,1) = \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$$

$$P_X(1) = \frac{1}{2} \quad P_Y(0) = P_Y(1) = \frac{1}{2}$$

$$\text{a) } E[X] = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 1 = \frac{1}{2} \quad E[Y] = \frac{1}{2}$$

$$\text{b) } \text{Var}(X) = E[X^2] - (E[X])^2 = \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$E[X^2] = \frac{1}{2} \cdot 0^2 + \frac{1}{2} \cdot 1^2 = \frac{1}{2}$$

$$\text{Var}(Y) = \frac{1}{4}$$

$$c) \operatorname{Cov}(X, Y) = E[XY] - E[X]E[Y] = \frac{3}{8} - \frac{1}{2} \cdot \frac{1}{2}$$

$$\left[E[XY] = 0 \cdot 0 \cdot \frac{3}{8} + 0 \cdot 1 \cdot \frac{1}{8} + 1 \cdot 0 \cdot \frac{1}{8} + 1 \cdot 1 \cdot \frac{3}{8} = \frac{3}{8} \right]$$

$$\Rightarrow \operatorname{Cov}(X, Y) = \frac{3-2}{8} = \frac{1}{8}$$

$$d) \operatorname{Var}(2X+3Y) = 4 \operatorname{Var}(X) + 9 \operatorname{Var}(Y) + 2 \cdot 3 \cdot 2 \operatorname{Cov}(X, Y)$$

$$= 4 \cdot \frac{1}{4} + 9 \cdot \frac{1}{4} + \overset{6}{\cancel{12}} \cdot \frac{1}{\cancel{8}_4} = \frac{4+9+6}{4} = \frac{19}{4}$$